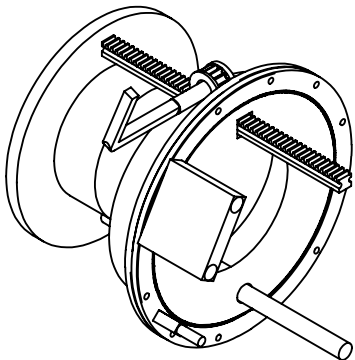
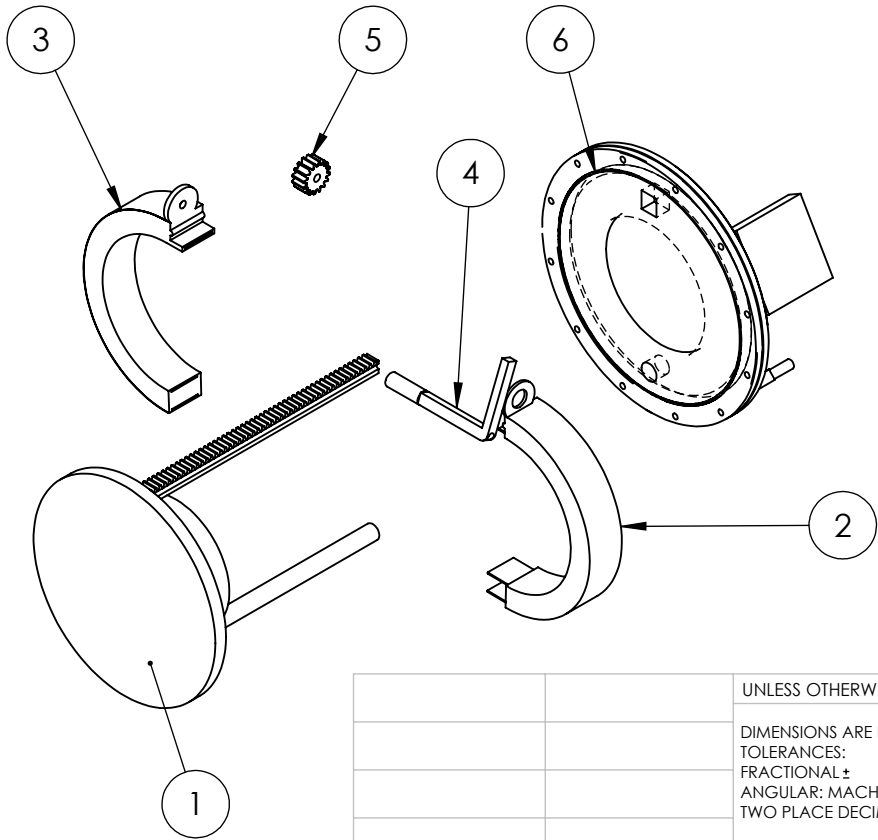


B

A

ITEM NO.	PART NUMBER	DESCRIPTION	QTY.
1	Base		1
2	Top Casing		1
3	Bottom Casing		1
4	Handle		1
5	Pinion		1
6	Slew Bearing		1



B

A

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		DIMENSIONS ARE IN INCHES TOLERANCES: FRACTIONAL ± ANGULAR: MACH ± BEND ± TWO PLACE DECIMAL ±0.5		DRAWN	
		INTERPRET GEOMETRIC TOLERANCING PER:		CHECKED	
		MATERIAL PLALA		ENG APPR.	
		FINISH		MFG APPR.	
				Q.A.	
				COMMENTS:	
NEXT ASSY	USED ON				
APPLICATION		DO NOT SCALE DRAWING			

TITLE:

E4Assembly

SIZE

A

DWG. NO.

REV

SCALE: 1:8

WEIGHT:

SHEET 1 OF 12

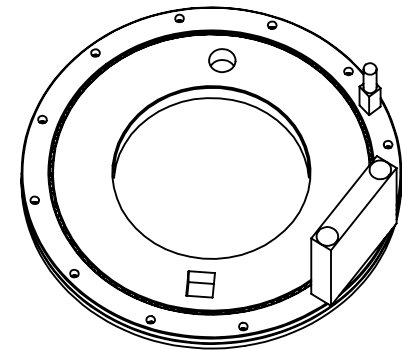
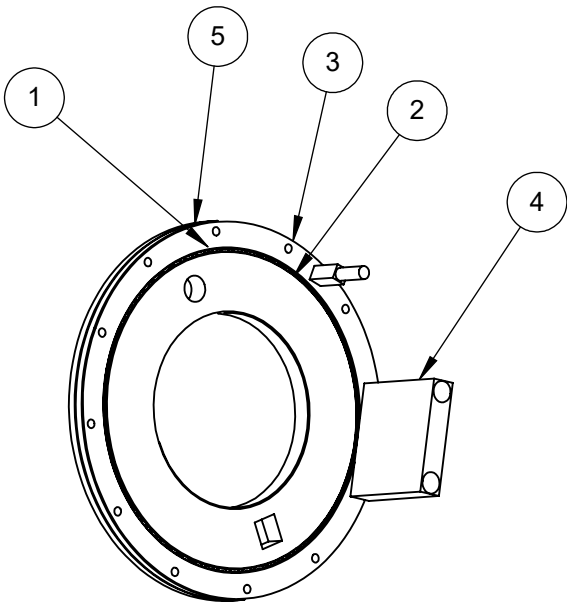
2

1

ITEM NO.	PART NUMBER	DESCRIPTION	QTY.
1	Bead		64
2	Inner Rail		1
3	Outer Rail w/Handle		1
4	Wall		1
5	Outer Rail		1

B

B



A

A

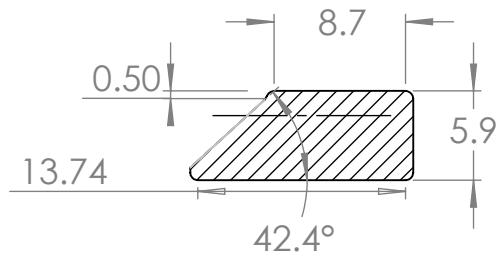
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		UNLESS OTHERWISE SPECIFIED:		NAME	DATE	TITLE: Slew Bearing		
		DIMENSIONS ARE IN INCHES	DRAWN					
		TOLERANCES:	CHECKED					
		FRACTIONAL ±	ENG APPR.					
		ANGULAR: MACH ± BEND ±	MFG APPR.					
		TWO PLACE DECIMAL ±0.5	Q.A.			SIZE A DWG. NO. REV		
		INTERPRET GEOMETRIC TOLERANCING PER:	COMMENTS:					
		MATERIAL						
		FINISH						
NEXT ASSY	USED ON							
APPLICATION		DO NOT SCALE DRAWING				SCALE: 1:8	WEIGHT:	SHEET 2 OF 12

2

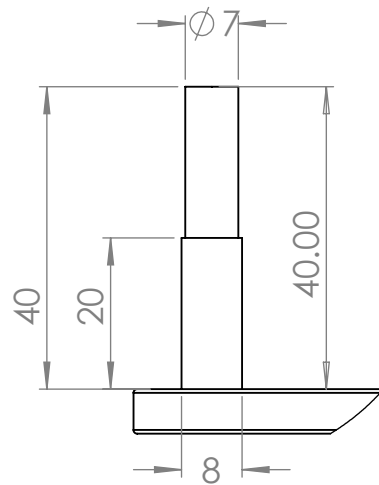
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B



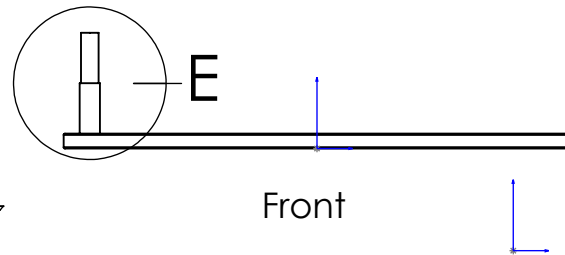
DETAIL A

SCALE 2 : 1

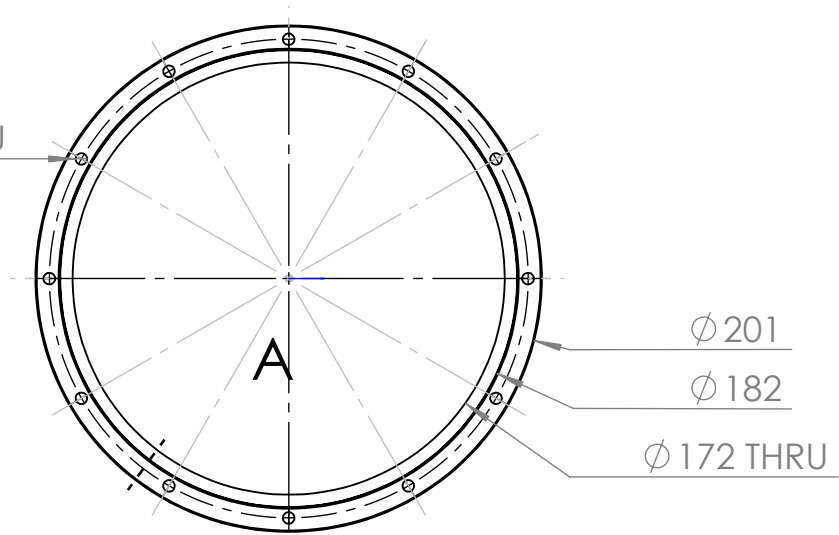


DETAIL E

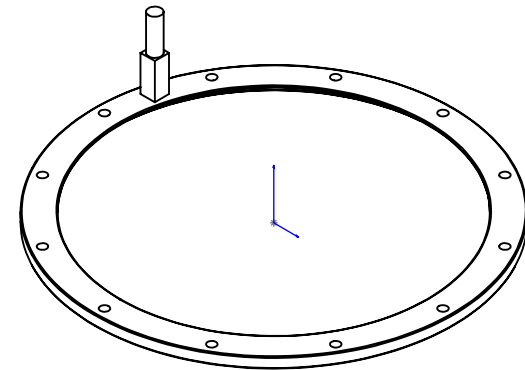
SCALE 1 : 1



Front

12X $\varnothing 4.6$ THRU

Bottom



B

A

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		UNLESS OTHERWISE SPECIFIED: DIMENSIONS ARE IN INCHES TOLERANCES: FRACTIONAL ± ANGULAR: MACH ± BEND ± TWO PLACE DECIMAL ±0.5 DEBURR ALL EDGES	DRAWN	NAME	DATE	TITLE: Outer Rail w/Handle		
			CHECKED					
			ENG APPR.					
			MFG APPR.					
			Q.A.					
		INTERPRET GEOMETRIC TOLERANCING PER:	COMMENTS:			SIZE	DWG. NO.	REV
		MATERIAL PLA				A	1	
NEXT ASSY	USED ON	FINISH						
APPLICATION		DO NOT SCALE DRAWING				SCALE 1:3		SHEET 3 OF 12

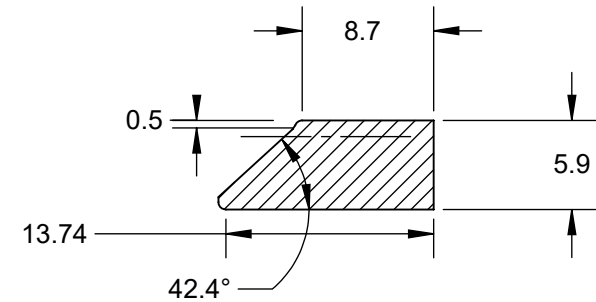
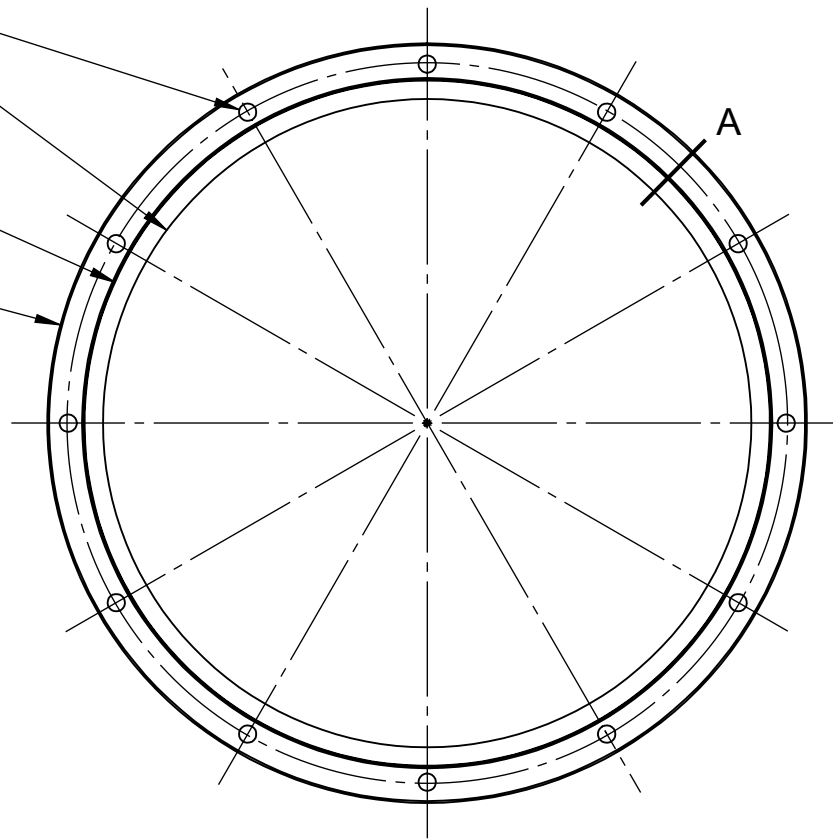
B

12X Ø4.6 THRU

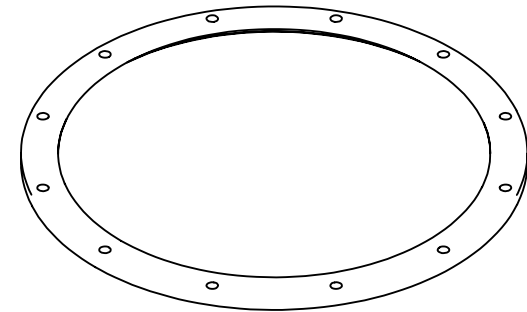
Ø171.5 THRU

Ø181.3

Ø201



DETAIL VIEW A
SCALE 1:1



B

A

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NEXT ASSY

USED ON

APPLICATION

UNLESS OTHERWISE SPECIFIED:

DIMENSIONS ARE IN INCHES
TOLERANCES:
FRACTIONAL ±
ANGULAR: MACH ± BEND ±
TWO PLACE DECIMAL ±0.5
DEBURR ALL EDGES

INTERPRET GEOMETRIC
TOLERANCING PER:

MATERIAL

PLA

FINISH

DO NOT SCALE DRAWING

NAME

DATE

DRAWN

CHECKED

ENG APPR.

MFG APPR.

Q.A.

COMMENTS:

TITLE:

Outer Rail

SIZE

A

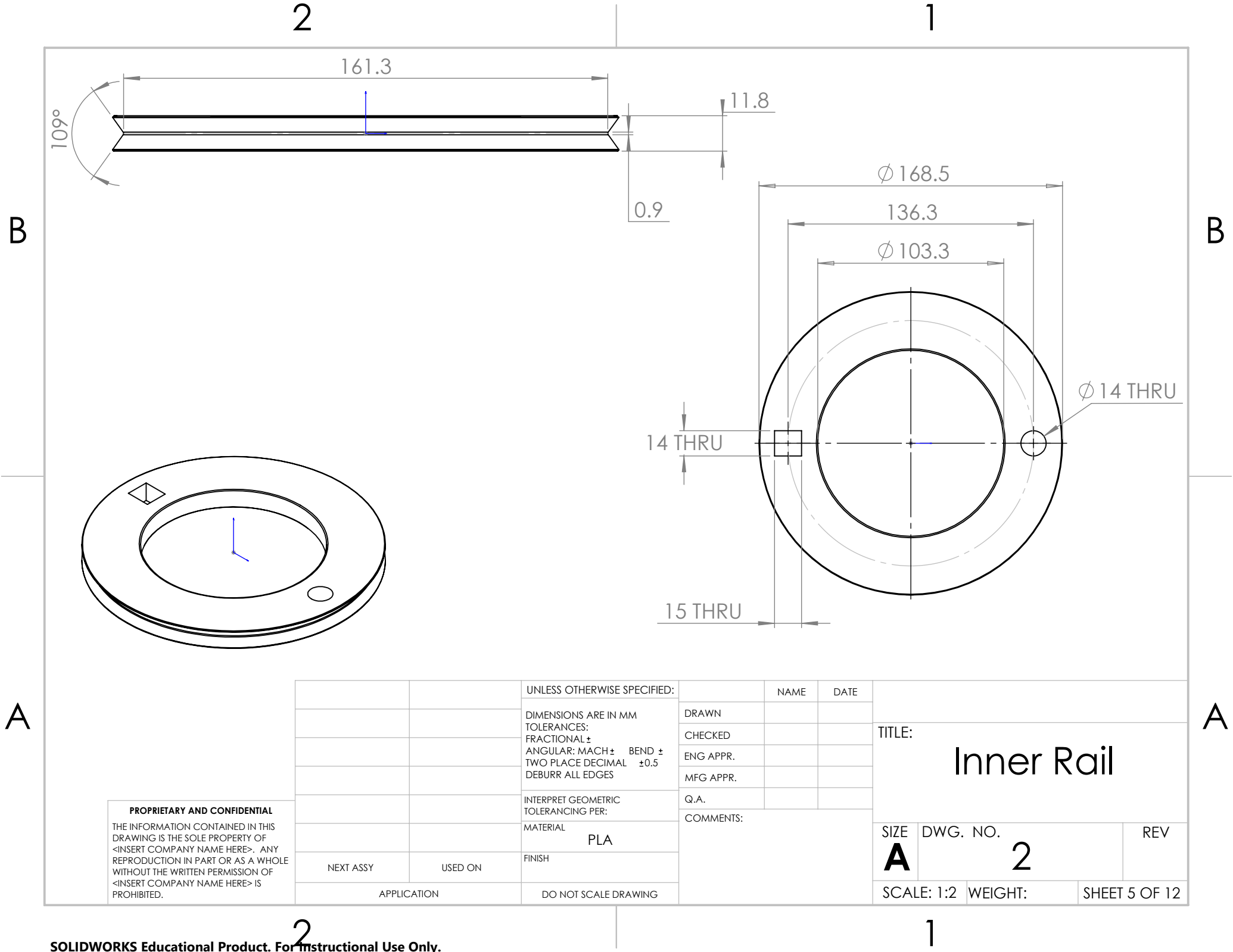
DWG. NO.

2

REV

SCALE: 1:1

SHEET 4 OF 12



A

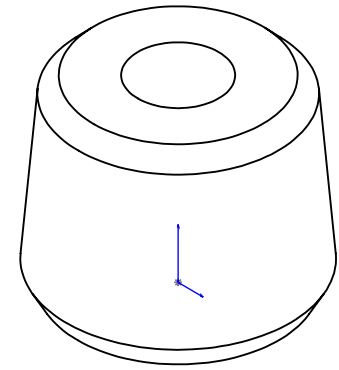
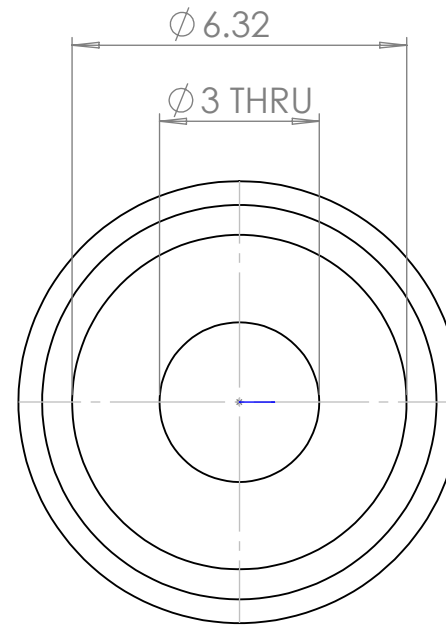
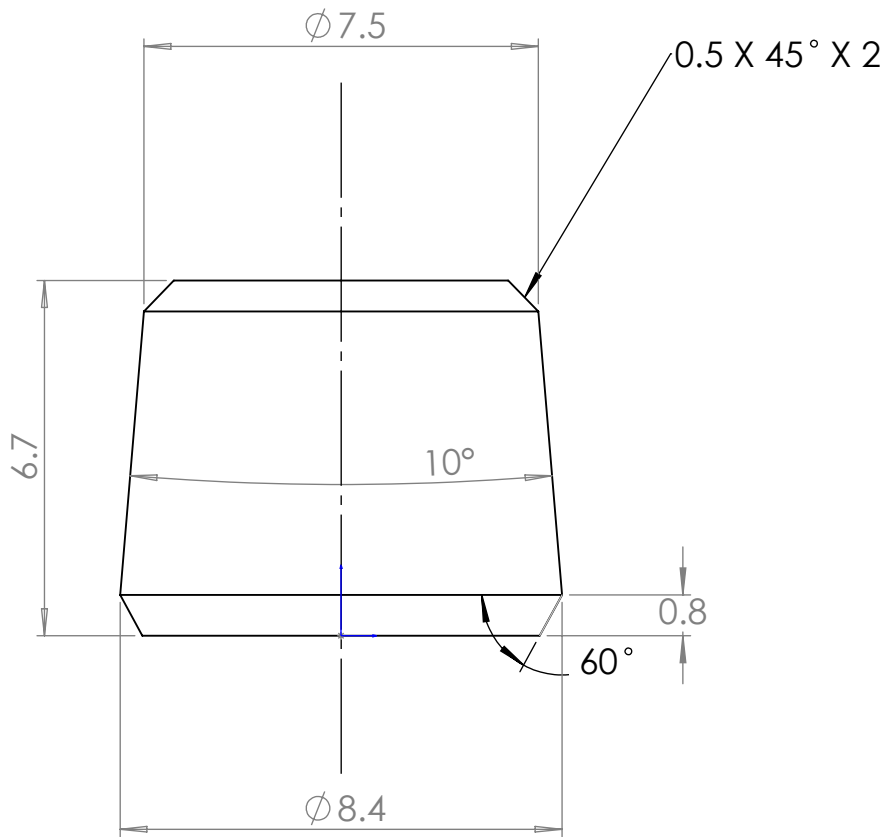
A

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		DIMENSIONS ARE IN MM	DRAWN		
		TOLERANCES:	CHECKED		
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		ANGULAR: MACH ± BEND ±	MFG APPR.		
		TWO PLACE DECIMAL ±0.5	Q.A.		
		DEBURR ALL EDGES	COMMENTS:		
		INTERPRET GEOMETRIC TOLERANCING PER:			
		MATERIAL PLA			
NEXT ASSY	USED ON	FINISH			
	APPLICATION	DO NOT SCALE DRAWING			

TITLE:		
Inner Rail		
SIZE	DWG. NO.	REV
A	2	
SCALE: 1:2	WEIGHT:	SHEET 5 OF 12

B



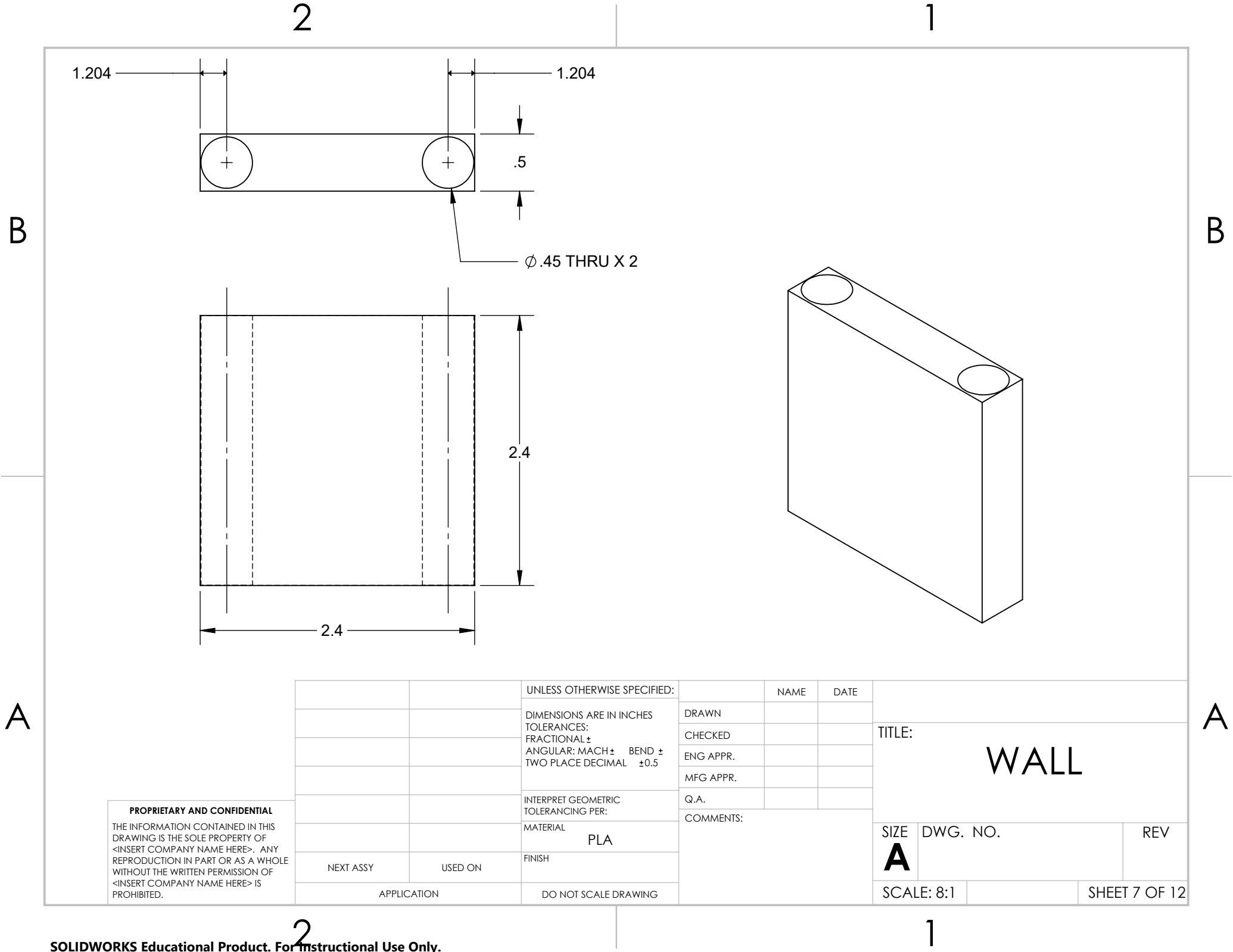
B

A

A

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		UNLESS OTHERWISE SPECIFIED:		NAME	DATE	TITLE: Rail Bead			
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		FRACTIONAL ±	ENG APPR.						
		ANGULAR: MACH± BEND ±	MFG APPR.						
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B

B

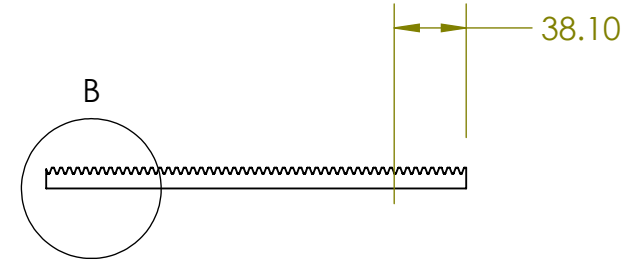
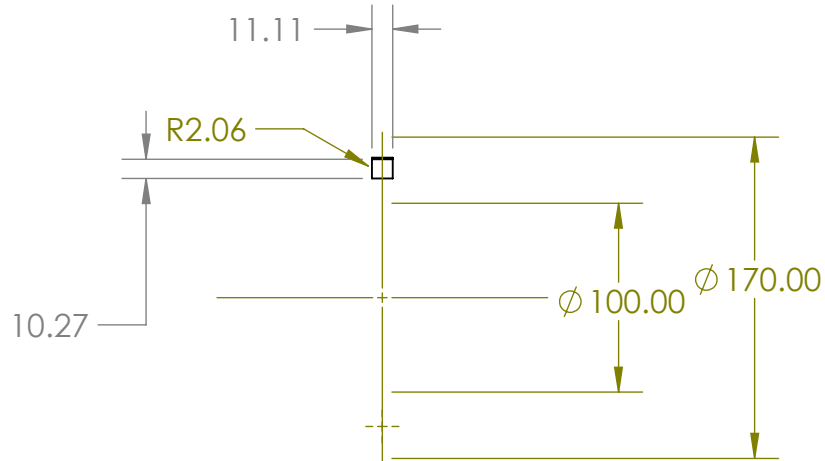
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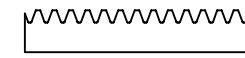
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		ANGULAR: MACH ± BEND ±	MFG APPR.					
		TWO PLACE DECIMAL ±0.5	Q.A.			SIZE A DWG. NO. REV		
		INTERPRET GEOMETRIC TOLERANCING PER:	COMMENTS:					
		MATERIAL PLA						
		FINISH						
NEXT ASSY	USED ON					SCALE: 8:1 SHEET 7 OF 12		
APPLICATION		DO NOT SCALE DRAWING						

B



B

McMaster Rack:
Pitch Height: 0.11"
Gear Pitch: 64
Face Width: 1/8"



DETAIL B
SCALE 1 : 2

Note: Scaled by factor of 3.5

A

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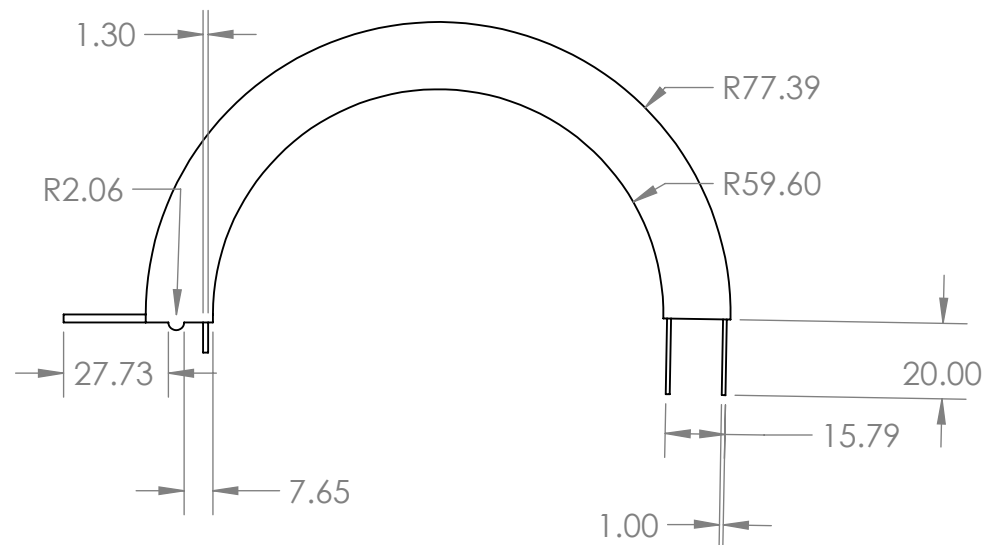
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			CHECKED					
			ENG APPR.					
			MFG APPR.					
		INTERPRET GEOMETRIC TOLERANCING PER:	Q.A.			SIZE A DWG. NO. REV		
		MATERIAL PLA	COMMENTS:					
		FINISH						
NEXT ASSY	USED ON							
APPLICATION		DO NOT SCALE DRAWING	SCALE: 1:4				WEIGHT:	SHEET 8 OF 12

2

1

1

B



A

UNLESS OTHERWISE SPECIFIED:

NAME	DATE
------	------

DRAWN

CHECKED

ENG APPR.

Q.A.

COMMENTS:

TITLE:

Top Casing

SIZE	PRICE	SALES
Small	10	100
Medium	20	200
Large	30	300

DWG. NO.	
----------	--

REV

SCALE: 1:4

WEIGHT:

SHEET 9 OF 12

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DIMENSIONS ARE IN INCHES
TOLERANCES:
FRACTIONAL $\pm$
ANGULAR: MACH $\pm$ BEND $\pm$
TWO PLACE DECIMAL ± 0.5

INTERPRET GEOMETRIC
TOLERANCING PER:

MATERIAL	PLA
----------	-----

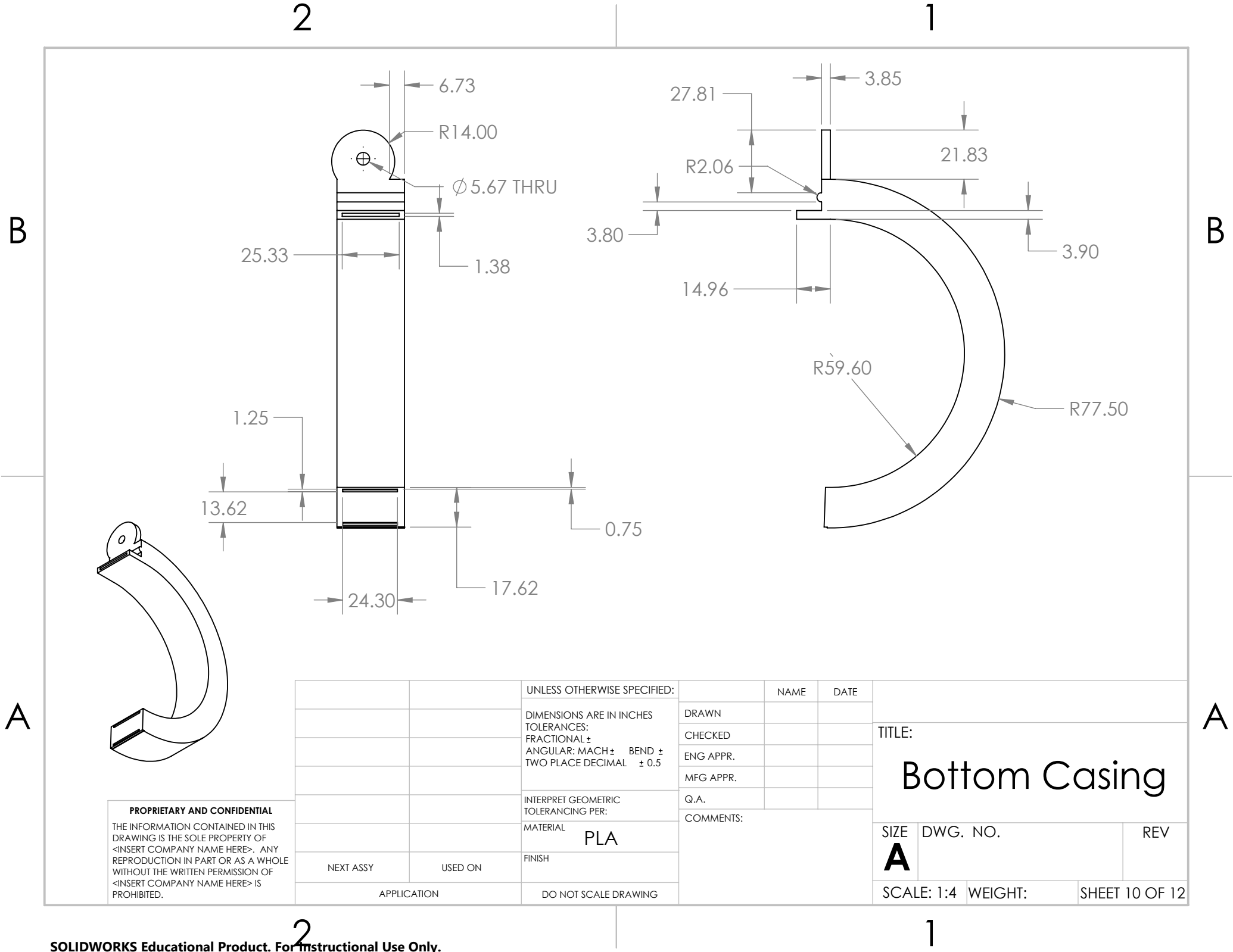
FINISH

NEXT ASSY

USED ON

APPLICATION

DO NOT SCALE DRAWING

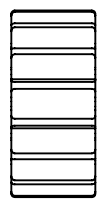
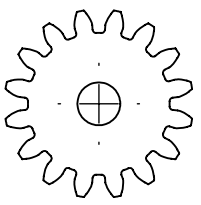


2

1

B

B



McMaster-Carr Gear:
Gear Pitch: 64
Number of Teeth: 16
Gear Pitch Diameter: 1/4"
Face Width: 1/8"
For Shaft Diameter: 1/8"

Note: Scaled at factor of 3.5

A

A

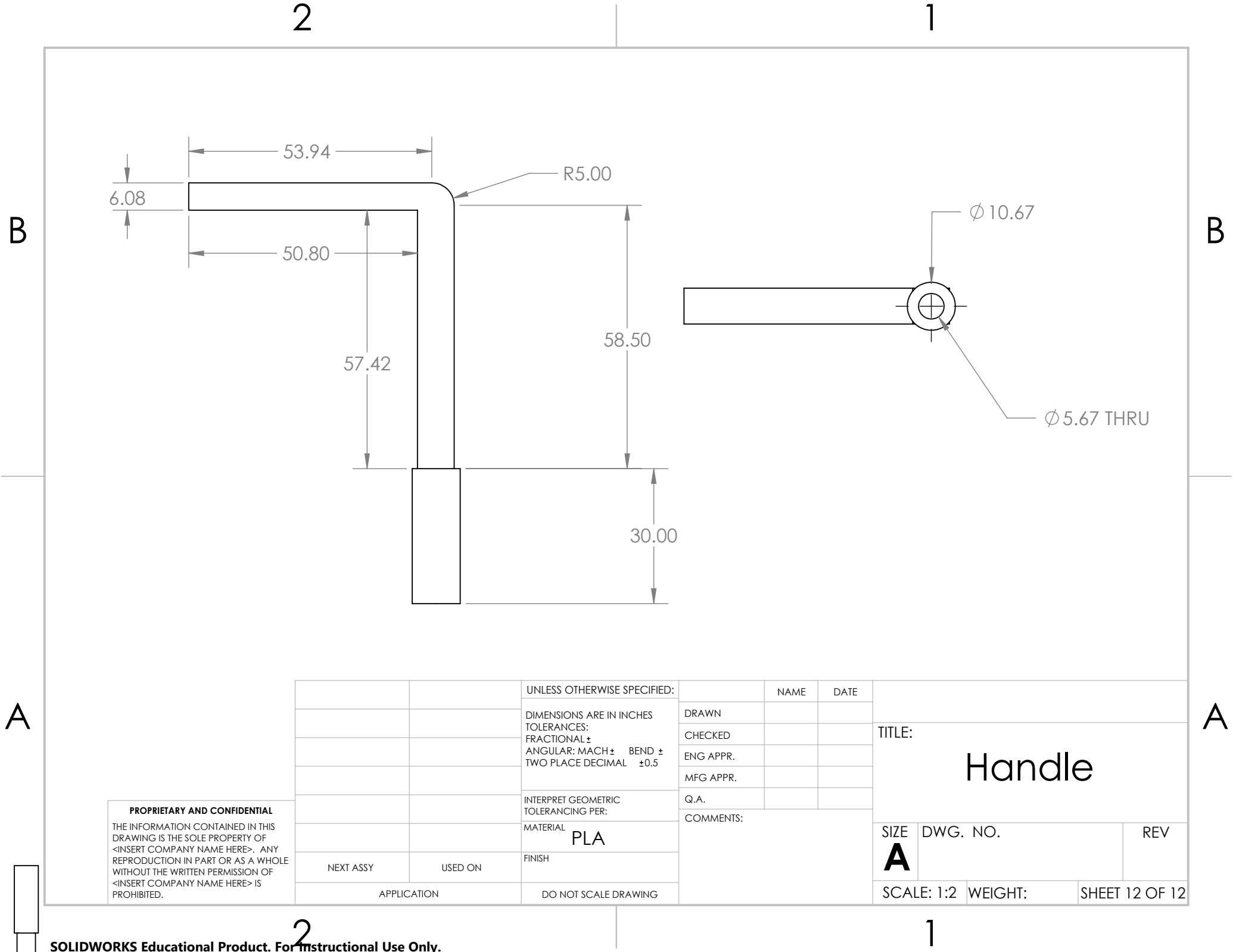
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		UNLESS OTHERWISE SPECIFIED:		NAME	DATE
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		TOLERANCES:	CHECKED		
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		ANGULAR: MACH ± BEND ±	MFG APPR.		
		TWO PLACE DECIMAL ±0.5	Q.A.		
		INTERPRET GEOMETRIC	COMMENTS:		
		TOLERANCING PER:			
		MATERIAL PLA			
		FINISH			
NEXT ASSY	USED ON				
APPLICATION		DO NOT SCALE DRAWING			

TITLE: Pinion		
SCALE: 1:1	WEIGHT:	SHEET 11 OF 12

2

1



A

A

B

B

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		DIMENSIONS ARE IN INCHES
		TOLERANCES:
		FRACTIONAL ±
		ANGULAR: MACH ± BEND ±
		TWO PLACE DECIMAL ±0.5
		INTERPRET GEOMETRIC TOLERANCING PER:
		MATERIAL PLA
NEXT ASSY	USED ON	FINISH
APPLICATION		DO NOT SCALE DRAWING

	NAME	DATE
DRAWN		
CHECKED		
ENG APPR.		
MFG APPR.		
Q.A.		
COMMENTS:		

TITLE:		
Handle		
SIZE	DWG. NO.	REV
A		
SCALE: 1:2	WEIGHT:	SHEET 12 OF 12

Bill of Materials

Item	Part Number (ASIN/SKU)	Manufacturer	Quantity	Price per unit (\$)	Total cost for part (\$)	Notes
3.3V Raspberry Pi laser	99634402	PMD Way	1	1.50	\$1.50	The laser comes as part of the Raspberry pi sensor pack. Here is the link to the product
AA battery	B07MLFBJG3	Amazon	2	0.39	\$0.78	Comes in a pack of 40, however only 2 are needed for the battery casing and the protot
Battery Casing	B00SXA8KPG	EX Electronix Expr	1	2.99	\$2.99	The casing that holds 2 AA batteiers, which are needed to operate the laser. Here is the
On/Off switch	B07S2QJKTX	DaierTek	1	0.80	\$0.80	Comes in a pack of 10, however only 1 is needed to operate the laser. Here is the link t
Hex Nut Zinc #8-32	1006540053	Home Depot	4	0.13	\$0.52	Comes in a pack of 12, but only 4 are needed to stabilize the threaded rod with respect
#8-32 x 1 ft Zinc Plated Steel (Coarse) Threaded Rod	670960	Home Depot	1	1.36	\$1.36	Cut down the threaded rod to 6 inches with the use of pliers. Here is the link to the prod
1mm CS X 10mm ID (NBR) Buna-N 70 Duro Metric O-Ring	N1_00X010	theoringstore	1	0.08	\$0.08	2 Silicon O Rings are needed to induce friction with respect to the handle. Here is the lin
Rubber Washer, 5/16" Inner Diamter, 3/4" Outer Diameter, 1/16" Thick	B07G6SN5ZW	Primal23 Industrial	3	0.28	\$0.84	3 Rubber Washers are used to induce friction between the threaded rod and the nuts. H
DIN912, Socket Cap Head, Alloy Steel, 14.9 Grade & M3-0.5 Hex Socket Screws, Thread Diamter 3mm, Pitch 0.5 mm, Thread Length 16mm	B09MYDQCXX	VIGRUE	1	17.99	\$17.99	12 M3 screws and nuts are needed to secure the slew bearing. Here is the link to the pr
Nylon Filament	B087R3M9Z2	Micro Center	1	30.59	\$30.59	The filament is needed in order to 3d print our design. Here is the link to the product htt
2P-10 Medium Viscosity, 2.25 O	2P-10 MED 2 OZ	Fast Cap	1	10.65	\$10.65	This product is the general advesive product that was used to glue the pinion to the han
Total cost for prototype:				\$57.45		

HEIGHT GAUGE PROCESS ROUTER

Circular Bearing

Process	Machine/Tool	Comments
3D-Print the files Wall.STL, inner rail.STL, outer rail w/ handle.STL, outer rail.STL, and Bead.STL	3D-Printer	Print out a set of 64 beads from the bead STL for future assembly.
Assemble the inner and outer parts of the rail by inserting the beads and screwing	Alan Wrench	Start with the inner rail and only race the outer rail without the handle and slip in beads till the rail is filled. Then place the race outer with the handle part on top of the other race outer part. Lastly, align the holes for the outer rails and place screws (MORE INFO ON SCREW)
Insert the wall 3D-Print into the screws	N/A	The sizing for the Wall.STL should allow the holes to fit onto the specific screws on the outer rail. Keep in mind the laser will be parallel to the wall.

Base

Process	Machine/Tool	Comments
3D-Print E4Base.STL	3D-Print (Powdered Nylon)	Print the base in a horizontal orientation to increase durability in the xy plane

Top/Bottom Gear Casing and Circle Handle

Process	Machine/Tool	Comments
3D-Print the files (E4CircleHandle.STL,E4NewBottom.STL, and E4RectangularTop.STL)	3D-Printer	Print handle horizontally for the most durable handle. Print others in suggested optimal position
Glue handle to the gear to make the full handle	2P-10 or Super Glue	Make sure to measure the radius of the handle and draw a cutout where to glue the handle on the gear.
As you assemble the pieces, place two silicon o-rings, one on the inside of where the handle and casing connect, and one on the outside	N/A	Place an o-ring on the handle and then put it through the hole on one of the casings. Place a second o-ring on the handle that will sit outside the casing hole for now. Then connect the two casings. Lastly, push the o-rings into the hole till secure.
Add threaded rod and double nut	Plyer	Threaded Rod Zinc 6X8-32 that goes through the system, and 2 Hex Nut Zinc #8-32 on each side to tighten the system. Utilize a pair of pliers to increase tightness.
Add assembled piece to base print	N/A	Make sure you place the assembled casing with the fully flat side up. The rounded hole part sticks out on one side, so the flat side should be where the rail goes.

Threaded Rod in Handle

Process	Machine/Tool	Comments
Cut bought threaded rod to 6"	Horizontal Bandsaw	Use the horizontal bandsaw or any other cutting technique to cut down the 1 ft long rod. Make sure to wrap in a rag to keep from damaging the thread.
Sand edges of the rod	Sandpaper	Sand down the cut edge of the rod until it is smooth to prevent potential injuries

Laser Electronics

Process	Machine/Tool	Comments
Strip wires to the 3.3V Raspberry Pi Laser and the Battery Casing	Wire Strippers	Strip the wires to both to create space for soldering. Make sure not to take too much wire casing off.
Solder the Raspberry Pi Laser, Battery Casing, and On/Off Switch together	Soldering Iron, Solder Wire, Stand, Sponge	If needed, take extra wires for more space. Solder the ground wire of the laser and battery casing together. Then solder the positive wire from the battery casing to the center prong of the on/off switch. Finally, solder the positive wire of the laser to the outer prong of the on/off laser. If confused on anything, wrap wires to test before soldering.
Glue sides of the laser to the circular bearing outside section	2P-10 or Superglue	Be sure to place glue carefully in order to not block the laser or connection to the wires. Also be sure the laser is parallel to the wall connected on the circular bearing.